

AIE-CASE - Color-Invariant Saree Design Recognition

1. Objective

Build and evaluate a system that identifies a saree by the design on its surface, independent of the color palette in which that design is rendered.

Think of it as face recognition for textiles: given a query image, the system must match it against a reference database of known saree designs. The current scope is sarees only; the approach should plausibly extend to other garment categories later.

2. Task Definition

- Input: an RGB image of a saree.
- Reference database (gallery): images of known designs, where the same design may appear in multiple colorways.
- Output:
 - Identification — rank the gallery by similarity to the query and return the best match(es).
 - Verification — given a pair of images, decide whether they carry the same design.
 - Core requirement — color invariance: the same motif in different palettes must match; different motifs must not match, even when rendered in identical palettes.

3. Deliverables

1. **Approach note (500 characters):** chosen architecture and the rationale behind it; the complete pre- and post-processing pipeline; training strategy (loss functions, sampling, augmentations).
2. **Working code:** an end-to-end PyTorch implementation covering training and inference, runnable as-is (Kaggle Notebook or repository).
3. **Evaluation:** propose and justify your evaluation protocol, and report results against it. Strong submissions typically cover both identification and verification, with a clearly documented gallery/query split.
4. **Efficiency report (bonus):** parameter count, FLOPs and/or inference latency, embedding size. Additional credit for lean, fast, or unconventional architectures — provided you can defend every choice.

4. Datasets

1. DeepLure saree corpus:
https://drive.google.com/open?id=1V_DcXJ50QYV9nEqvq7fqKno_lhbSPdBM
2. Indian Saree Patterns (Kaggle): <https://www.kaggle.com/datasets/div456/indian-saree-patterns>

You may combine, re-split, clean, or relabel the data as needed — document whatever you do. The Drive corpus is proprietary to 3rd party vendors: do not redistribute it and delete your copy once the exercise concludes.

5. Rules

- **Framework:** PyTorch.
- **Compute:** Kaggle Notebooks are permitted (the free GPU tier is sufficient).
- **Pretrained backbones** and public checkpoints are allowed; disclose everything you use.
- **External data** is allowed with disclosure.
- All work must be your own. Post submission we'll conduct a live follow-up discussion and should be able to explain and defend every decision from memory.

6. Submission

- **Deadline:** 11:59 PM IST, 5th October 2026.
- **Submit:** the notebook/repository link at <https://forms.gle/1fJaAFbuTS4FdhQn9>
- Ensure all links are viewable by the reviewer before the deadline.

7. What We Assess

Problem formulation, soundness and rigor of the evaluation, code quality and reproducibility, clarity of reasoning, and the efficiency/novelty of the architecture.